

Conditional Probability: So far, we have discussed about the probabilities of events based on the assumption that no information was available about the random experiment other than the sample space. Sometimes, however, it is known that an event E_2 has occurred. The question is, can we use this information in making a statement about the outcome of another event E_1 ? How?

Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $E_2 \in \mathcal{F}$ with $P(E_2) > 0$. Then, for an arbitrary $E_1 \in \mathcal{F}$ we shall write

$$P(E_1|E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)}, \quad (1)$$

and the quantity so defined is known as the *conditional probability* of E_1 , given E_2 .

Remark 1: Conditional probability remains undefined when $P(E_2) = 0$.

Theorem 1: Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $H \in \mathcal{F}$ with $P(H) > 0$. Then, $(\Omega, \mathcal{F}, P_H)$, where $P_H(A) = P\{A|H\} \forall A \in \mathcal{F}$, is a probability space.

Proof: Clearly, $P_H(A) = P\{A|H\} \geq 0, \forall A \in \mathcal{F}$. Also,

$$P_H(\Omega) = \frac{P(\Omega \cap H)}{P(H)} = 1.$$

If $A_1, A_2, \dots$ is a disjoint sequence of sets in $\mathcal{F}$, then

$$\begin{aligned} P\left(\bigcup_{i=1}^{\infty} A_i | H\right) &= \frac{P\left\{\bigcup_{i=1}^{\infty} A_i \cap H\right\}}{P(H)} \\ &= \frac{P\left\{\bigcup_{i=1}^{\infty} (A_i \cap H)\right\}}{P(H)}. \end{aligned} \quad (2)$$

Since A_i are disjoint, $A_i \cap H$ are also disjoint. Thus we shall have

$$\begin{aligned} \frac{P\left\{\bigcup_{i=1}^{\infty} (A_i \cap H)\right\}}{P(H)} &= \sum_{i=1}^{\infty} \frac{P(A_i \cap H)}{P(H)} \\ &= \sum_{i=1}^{\infty} P(A_i | H). \end{aligned} \quad (3)$$

Substituting (3) in (2), we obtain

$$P\left(\bigcup_{i=1}^{\infty} A_i | H\right) = \sum_{i=1}^{\infty} P(A_i | H).$$

One can observe that $P_H(A)$ is a probability measure on $(\Omega, \mathcal{F})$ as it satisfies all three axioms of a probability function. Also $(\Omega, \mathcal{F}, P_H)$ will be a probability space.

Properties of Conditional Probability:

1. **Multiplication Rule:** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A, B \in \mathcal{F}$, with $P(A) > 0, P(B) > 0$. Then

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B).$$

2. **General Multiplication Rule:** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $A_1, A_2, \dots, A_n \in \mathcal{F}$, with $P\left(\bigcap_{i=1}^{n-1} A_i\right) > 0$. Then

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P\left\{A_n \left| \bigcap_{i=1}^{n-1} A_i \right.\right\}.$$

3. **Law of Total Probability:** Let $A \in \mathcal{F}$ and let $H_i, i = 1, 2, \dots$ be events that partition Ω , (i.e., $\bigcup_{i=1}^{\infty} H_i = \Omega$ and $H_i \cap H_j = \phi \ \forall \ i \neq j$) with $P(H_i) > 0, \ \forall \ i$. Then,

$$P(A) = \sum_{i=1}^{\infty} P(A|H_i)P(H_i).$$

Proof: Since $H_i, i = 1, 2, \dots$ partition Ω , thus $A \cap H_i$ partition A . Hence, using the axiom of countable additivity, we have

$$P(A) = P\left\{\bigcup_{i=1}^{\infty} (A \cap H_i)\right\} = \sum_{i=1}^{\infty} P(A \cap H_i),$$

and

$$P(A \cap H_i) = P(A|H_i)P(H_i).$$

This implies

$$\sum_{i=1}^{\infty} P(A \cap H_i) = \sum_{i=1}^{\infty} P(A|H_i)P(H_i).$$

The result now follows on combining above expressions.

Note: In particular, if H is such that $0 < P(H) < 1$, we have

$$P(A) = P(A|H)P(H) + P(A|H^c)P(H^c).$$

4. **Bayes Theorem:** Let H_i be a disjoint sequence of events such that $P(H_i) > 0, i = 1, 2, \dots$, and $\bigcup_{i=1}^{\infty} H_i = \Omega$. Let $B \in \mathcal{F}$ with $P(B) > 0$. Then

$$P(H_i|B) = \frac{P(H_i)P(B|H_i)}{\sum_{i=1}^{\infty} P(H_i)P(B|H_i)}, \quad i = 1, 2, \dots$$

Proof: Using multiplication rule, we have

$$P(B \cap H_i) = P(B)P(H_i|B) = P(H_i)P(B|H_i),$$

and it follows that

$$P(H_i|B) = \frac{P(H_i)P(B|H_i)}{P(B)}.$$

Using the law of total probability, it can be written as

$$P(H_i|B) = \frac{P(H_i)P(B|H_i)}{\sum_{i=1}^{\infty} P(H_i)P(B|H_i)},$$

and the theorem follows.

Remark 1: Let $H_1, H_2, \dots$ are all the “causes” that lead to the outcome of a random experiment. Let H_i be the set of outcomes corresponding to the i^{th} cause. Assume that the probabilities $P(H_i), i = 1, 2, \dots$ called the *prior probabilities*, can be assigned. Now suppose that the experiment results in an event B of positive probability. This information leads to a reassessment of the prior probabilities. The conditional probabilities $P(H_i|B)$ are called the *posterior probabilities*. The Bayes rule can be interpreted as a rule giving the probability that the observed event B was due to cause H_j .

Example 1: An agency “A” reports on-time performance for airlines at major Indian airports. Air India, Indigo, and Spicejet share terminal 3 at IGI airport Delhi. Suppose that the percentage of on-time flights reported was 76.8% for Air India, 71.5% for Indigo, and 82.2% for Spicejet. Assume that 30% of the flights arriving at terminal 3 are Air India flights, 32% are Indigo flights, and 38% are Spicejet flights. An announcement is made that Flight 1382 will be arriving at gate no. 20 of terminal 3. What is the probability that Flight 1382 will arrive on time?

Solution: Let O = flight arrives on time

L = flight arrives late

A = Air India flight

I = Indigo flight

S = Spicejet flight

Given that

$$P(O|A) = 0.768, P(O|I) = 0.715, P(O|S) = 0.822,$$

$$P(A) = 0.30, P(I) = 0.32, P(S) = 0.38.$$

Using multiplication law, we have

$$P(A \cap O) = P(O|A)P(A) = 0.768 \times 0.30 = 0.2304,$$

$$P(I \cap O) = P(O|I)P(I) = 0.715 \times 0.32 = 0.2288,$$

$$P(S \cap O) = P(O|S)P(S) = 0.822 \times 0.38 = 0.3124.$$

Now the probability of an on-time flight is given by

$$P(O) = 0.2304 + 0.2288 + 0.3124 = 0.7716.$$

Example 2: Black boxes used in aircrafts are manufactured by three companies A, B and C. 75% are manufactured by A, 15% by B, and 10% by C. The defect rates of black boxes manufactured by A, B, and C are 4%, 6%, and 8%, respectively. If a black box tested randomly is found to be defective, what is the probability that it is manufactured by company A?

Solution: Let $P(A)$, $P(B)$, $P(C)$ be the events corresponding to the black box being manufactured by companies A, B, and C, respectively, and $P(D)$ be the probability of a defective black box. We are interested in calculating the probability $P(A|D)$, which is given by

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)}.$$

Now we have $P(A) = 0.75$, $P(D|A) = 0.04$, and $P(D)$ is given by

$$\begin{aligned} P(D) &= P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C) \\ &= 0.75 \times 0.04 + 0.15 \times 0.06 + 0.10 \times 0.08 \\ &= 0.047. \end{aligned}$$

This implies that $P(A|D) = (0.04)(0.75)/(0.047) = 0.6382$.